

logo

|           |               |       |              |                                                                                         |    |
|-----------|---------------|-------|--------------|-----------------------------------------------------------------------------------------|----|
| Ncruc     | 58532and13532 | AgSp1 | nt           | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | rr 15        | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | sy           | ITRGFPDSPLEQPVVEEPQGGPQGGQAQYQPQGPSLGGGVQGTQQSPGGNLGAAAPSDPLGKAIQGKKREPSNNLGGILSDIFGGGD | 85 |
| Ncruc     | 58532and13532 | AgSp1 | rr 14        | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | rr 7         | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | rr 12        | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 93 | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 6 | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 1 | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 2 | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 3 | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 4 | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 5 | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | tr 1         | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | tr 3         | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 92 | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 91 | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 88 | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 89 | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 90 | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 87 | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | rr 1         | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | rr 11        | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | rr 2         | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | rr 6         | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | rr 9         | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | rr 3         | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | rr 13        | .....                                                                                   | 0  |
| Mlaby     | 7048          | AgSp1 | cr           | .....                                                                                   | 0  |
| Aarg      |               | AgSp1 | FC511 cr     | .....                                                                                   | 0  |
| Mhut      | 1271          | AgSp1 | cr           | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | ct           | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | rr 8         | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | rr 5         | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | rr 10        | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | rr 16        | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | rr 4         | .....                                                                                   | 0  |
| Ncruc     | 58532and13532 | AgSp1 | tr 2         | .....                                                                                   | 0  |
| consensus |               |       |              |                                                                                         |    |

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|           |               |       |              |                                                                                                                                                                       |     |
|-----------|---------------|-------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Ncruc     | 58532and13532 | AgSp1 | nt           | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 15        | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | sy           | DES YET YYY T Y E T E Y Y T T S Y E T Y E T E S Y E T G S Y E T G T F E T Y E T G S Y E T G T Y E T Y E T G S Y A T G T Y E T E S Y E T G S Y E T G T Y E T Y E T G S | 170 |
| Ncruc     | 58532and13532 | AgSp1 | rr 14        | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 7         | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 12        | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 93 | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 6 | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 1 | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 2 | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 3 | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 4 | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 5 | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | tr 1         | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | tr 3         | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 92 | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 91 | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 88 | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 89 | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 90 | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 87 | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 1         | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 11        | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 2         | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 6         | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 9         | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 3         | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 13        | .....                                                                                                                                                                 | 0   |
| Mlaby     | 7048          | AgSp1 | cr           | .....                                                                                                                                                                 | 0   |
| Aarg      |               | AgSp1 | FC511 cr     | .....                                                                                                                                                                 | 0   |
| Mhut      | 1271          | AgSp1 | cr           | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | ct           | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 8         | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 5         | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 10        | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 16        | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 4         | .....                                                                                                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | tr 2         | .....                                                                                                                                                                 | 0   |
| consensus |               |       |              |                                                                                                                                                                       |     |

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|           |               |       |              |                                                                                        |     |
|-----------|---------------|-------|--------------|----------------------------------------------------------------------------------------|-----|
| Ncruc     | 58532and13532 | AgSp1 | nt           | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 15        | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | sy           | YATGTYETESYETGSYETGTRHETGSYATGTYETESYETGTYETYETGSYATGTYETESYVVTGSYETYETGSYATGTCETESYVT | 255 |
| Ncruc     | 58532and13532 | AgSp1 | rr 14        | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 7         | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 12        | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 93 | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 6 | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 1 | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 2 | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 3 | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 4 | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 5 | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | tr 1         | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | tr 3         | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 92 | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 91 | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 88 | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 89 | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 90 | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 87 | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 1         | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 11        | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 2         | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 6         | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 9         | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 3         | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 13        | .....                                                                                  | 0   |
| Mlaby     | 7048          | AgSp1 | cr           | .....                                                                                  | 0   |
| Aarg      |               | AgSp1 | FC511 cr     | .....                                                                                  | 0   |
| Mhut      | 1271          | AgSp1 | cr           | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | ct           | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 8         | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 5         | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 10        | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 16        | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 4         | .....                                                                                  | 0   |
| Ncruc     | 58532and13532 | AgSp1 | tr 2         | .....                                                                                  | 0   |
| consensus |               |       |              |                                                                                        |     |

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|-----------|---------------|-------|--------------|---------------------------------------------------------------------------------------|-----|
| Ncruc     | 58532and13532 | AgSp1 | nt           | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 15        | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | sy           | GSYETGSYVTGETGSFETGSYVTGETGSFETGSYVTGESGSFETGSYVTGESGSFETGSYVTGESSSFETGSYVTGESGSFETGS | 340 |
| Ncruc     | 58532and13532 | AgSp1 | rr 14        | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 7         | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 12        | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 93 | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 6 | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 1 | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 2 | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 3 | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 4 | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 5 | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | tr 1         | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | tr 3         | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 92 | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 91 | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 88 | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 89 | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 90 | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 87 | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 1         | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 11        | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 2         | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 6         | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 9         | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 3         | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 13        | .....                                                                                 | 0   |
| Mlaby     | 7048          | AgSp1 | cr           | .....                                                                                 | 0   |
| Aarg      | AgSp1         | FC511 | cr           | .....                                                                                 | 0   |
| Mhut      | 1271          | AgSp1 | cr           | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | ct           | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 8         | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 5         | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 10        | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 16        | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 4         | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | tr 2         | .....                                                                                 | 0   |
| consensus |               |       |              |                                                                                       |     |

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|-----------|---------------|-------|--------------|--------------------------------------------------------------------------|-----|
| Ncruc     | 58532and13532 | AgSp1 | nt           | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 15        | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | sy           | YVTGESGSFETGSYVTGESGSFETGSYVTGESGSFETGSYVTGESGSFETGSYVTGESGSFETGSYVTGESG | 425 |
| Ncruc     | 58532and13532 | AgSp1 | rr 14        | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 7         | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 12        | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 93 | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 6 | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 1 | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 2 | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 3 | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 4 | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 5 | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | tr 1         | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | tr 3         | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 92 | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 91 | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 88 | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 89 | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 90 | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 87 | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 1         | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 11        | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 2         | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 6         | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 9         | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 3         | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 13        | .....                                                                    | 0   |
| Mlaby     | 7048          | AgSp1 | cr           | .....                                                                    | 0   |
| Aarg      | AgSp1         | FC511 | cr           | .....                                                                    | 0   |
| Mhut      | 1271          | AgSp1 | cr           | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | ct           | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 8         | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 5         | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 10        | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 16        | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 4         | .....                                                                    | 0   |
| Ncruc     | 58532and13532 | AgSp1 | tr 2         | .....                                                                    | 0   |
| consensus |               |       |              |                                                                          |     |

|           |               |       |              |                                                                                       |     |
|-----------|---------------|-------|--------------|---------------------------------------------------------------------------------------|-----|
| Ncruc     | 58532and13532 | AgSp1 | nt           | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 15        | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | sy           | SFETGSYVTGESGSFETGSYVTGESGSVETGSYVTGESGSFETGSYTTGESGSFTTGSYITGESGSFSTASYVSGDYYSYETGSY | 510 |
| Ncruc     | 58532and13532 | AgSp1 | rr 14        | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 7         | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 12        | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 93 | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 6 | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 1 | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 2 | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 3 | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 4 | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 5 | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | tr 1         | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | tr 3         | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 92 | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 91 | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 88 | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 89 | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 90 | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 87 | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 1         | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 11        | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 2         | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 6         | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 9         | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 3         | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 13        | .....                                                                                 | 0   |
| Mlaby     | 7048          | AgSp1 | cr           | .....                                                                                 | 0   |
| Aarg      | AgSp1         | FC511 | cr           | .....                                                                                 | 0   |
| Mhut      | 1271          | AgSp1 | cr           | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | ct           | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 8         | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 5         | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 10        | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 16        | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | rr 4         | .....                                                                                 | 0   |
| Ncruc     | 58532and13532 | AgSp1 | tr 2         | .....                                                                                 | 0   |
| consensus |               |       |              |                                                                                       |     |

Ncruc 58532and13532 AgSp1 nt  
Ncruc 58532and13532 AgSp1 rr 15  
Ncruc 58532and13532 AgSp1 sy  
Ncruc 58532and13532 AgSp1 rr 14  
Ncruc 58532and13532 AgSp1 rr 7  
Ncruc 58532and13532 AgSp1 rr 12  
Ncruc 58532and13532 AgSp1 serine rm 93  
Ncruc 58532and13532 AgSp1 glycine rm 6  
Ncruc 58532and13532 AgSp1 glycine rm 1  
Ncruc 58532and13532 AgSp1 glycine rm 2  
Ncruc 58532and13532 AgSp1 glycine rm 3  
Ncruc 58532and13532 AgSp1 glycine rm 4  
Ncruc 58532and13532 AgSp1 glycine rm 5  
Ncruc 58532and13532 AgSp1 tr 1  
Ncruc 58532and13532 AgSp1 tr 3  
Ncruc 58532and13532 AgSp1 serine rm 92  
Ncruc 58532and13532 AgSp1 serine rm 91  
Ncruc 58532and13532 AgSp1 serine rm 88  
Ncruc 58532and13532 AgSp1 serine rm 89  
Ncruc 58532and13532 AgSp1 serine rm 90  
Ncruc 58532and13532 AgSp1 serine rm 87  
Ncruc 58532and13532 AgSp1 rr 1  
Ncruc 58532and13532 AgSp1 rr 11  
Ncruc 58532and13532 AgSp1 rr 2  
Ncruc 58532and13532 AgSp1 rr 6  
Ncruc 58532and13532 AgSp1 rr 9  
Ncruc 58532and13532 AgSp1 rr 3  
Ncruc 58532and13532 AgSp1 rr 13  
MlabY 7048 AgSp1 cr  
Aarg AgSp1 FC511 cr  
Mhut 1271 AgSp1 cr  
Ncruc 58532and13532 AgSp1 ct  
Ncruc 58532and13532 AgSp1 rr 8  
Ncruc 58532and13532 AgSp1 rr 5  
Ncruc 58532and13532 AgSp1 rr 10  
Ncruc 58532and13532 AgSp1 rr 16  
Ncruc 58532and13532 AgSp1 rr 4  
Ncruc 58532and13532 AgSp1 tr 2  
consensus

[illegible]

Ncruc 58532and13532 AgSp1 nt  
Ncruc 58532and13532 AgSp1 rr 15  
Ncruc 58532and13532 AgSp1 sy  
Ncruc 58532and13532 AgSp1 rr 14  
Ncruc 58532and13532 AgSp1 rr 7  
Ncruc 58532and13532 AgSp1 rr 12  
Ncruc 58532and13532 AgSp1 serine rm 93  
Ncruc 58532and13532 AgSp1 glycine rm 6  
Ncruc 58532and13532 AgSp1 glycine rm 1  
Ncruc 58532and13532 AgSp1 glycine rm 2  
Ncruc 58532and13532 AgSp1 glycine rm 3  
Ncruc 58532and13532 AgSp1 glycine rm 4  
Ncruc 58532and13532 AgSp1 glycine rm 5  
Ncruc 58532and13532 AgSp1 tr 1  
Ncruc 58532and13532 AgSp1 tr 3  
Ncruc 58532and13532 AgSp1 serine rm 92  
Ncruc 58532and13532 AgSp1 serine rm 91  
Ncruc 58532and13532 AgSp1 serine rm 88  
Ncruc 58532and13532 AgSp1 serine rm 89  
Ncruc 58532and13532 AgSp1 serine rm 90  
Ncruc 58532and13532 AgSp1 serine rm 87  
Ncruc 58532and13532 AgSp1 rr 1  
Ncruc 58532and13532 AgSp1 rr 11  
Ncruc 58532and13532 AgSp1 rr 2  
Ncruc 58532and13532 AgSp1 rr 6  
Ncruc 58532and13532 AgSp1 rr 9  
Ncruc 58532and13532 AgSp1 rr 3  
Ncruc 58532and13532 AgSp1 rr 13  
MlabY 7048 AgSp1 cr  
Aarg AgSp1 FC511 cr  
Mhut 1271 AgSp1 cr  
Ncruc 58532and13532 AgSp1 ct  
Ncruc 58532and13532 AgSp1 rr 8  
Ncruc 58532and13532 AgSp1 rr 5  
Ncruc 58532and13532 AgSp1 rr 10  
Ncruc 58532and13532 AgSp1 rr 16  
Ncruc 58532and13532 AgSp1 rr 4  
Ncruc 58532and13532 AgSp1 tr 2  
consensus

| Sequence                                                                                                               | Position |
|------------------------------------------------------------------------------------------------------------------------|----------|
| . AYT . . . . . TPGKFRGPNKGPIHIIPAG . . . . .                                                                          | 45       |
| SGAV . . . . . PGIVPGPDGQPINIIPAG . . . . .                                                                            | 617      |
| PTTTT.KG . . . . . PG . . . . . SLLI . . . . .                                                                         | 36       |
| .SNT . . . . . TPGTVRGPDGQPIKIIPAG . . . . .                                                                           | 45       |
| .AAT . . . . . TPGSVKGPDGQPIQIEPAG . . . . .                                                                           | 45       |
| AFST . . . . . PGSIIPGPHGKPIHIGPAGPGTTTGAKLDSDDIS . . . . . EILLPSTP.SPPAP . . . . . Q . . . . . P                     | 76       |
| AFT . . . . . TPGSIIPGPDGQPIHVEPAGPGTTTGVQTGPDGK . . . . .                                                             | 59       |
| AFT . . . . . TPGSIIPGPDGQPIHVEPAGPGTTTGAETGPDGKVN . . . . . KIYLPSTP.KGPGG . . . . . P . . . . . G                    | 76       |
| AFT . . . . . TPGSIIPGPDGQPIHVEPAGPGTTTGAETGPDGKVN . . . . . KIYLPSTP.KGPGG . . . . . P . . . . . G                    | 76       |
| AFT . . . . . TPGSIIPGPDGQPIHVEPAGPGTTTGAETGPDGKVN . . . . . KIYLPSTP.KGPGG . . . . . P . . . . . G                    | 76       |
| AFT . . . . . TPGSIIPGPDGQPIHVEPAGPGTTTGAETGPDGKVN . . . . . KIYLPSTP.KGPGG . . . . . P . . . . . G                    | 76       |
| AFTT . . . . . PGSIIPGPDGRPIHVQPAG . . . . .                                                                           | 45       |
| AFST . . . . . PGTIIPGPDGKPIRVEPQS . . . . .                                                                           | 45       |
| AFST . . . . . PGTIIPGPHGKPIHIGPAGPGTTTGAK.DSDGSID . . . . . SIVLPSTP.KGSG . . . . . P . . . . . G                     | 74       |
| AFTT . . . . . PGTIKPHGKPIHVGPAAGPGTTTGAKTDSDDSID . . . . . SIVLPSTP.KGSG . . . . . P . . . . . G                      | 75       |
| AFTT . . . . . PGTIKPHGKPIHVKPAGPGTTTGAKTDSDDSID . . . . . SIVLPSTP.K . . . . . GSGP . . . . . G                       | 75       |
| AFTT . . . . . PGTIKPHGKPIHVKPAGPGTTTGAKTNSDDSID . . . . . SIVLPSTP.K . . . . . GSGP . . . . . G                       | 75       |
| AFTT . . . . . PGTIKPHGKPIHVRPAGPGTTTGAKTDSDDSID . . . . . SIVLPSTP.KSGPGFPTTPKSGSP . . . . . G                        | 86       |
| GPVFDF . . . . . HTSPSPITKPDGKPIQIVPAG . . . . .                                                                       | 81       |
| PSPRS.D.EP . . . . . KFVGKGVQPVKVVTA . . . . .                                                                         | 48       |
| GPRG . . . . . GV . . . . . PQGPGKKPIQILRG . . . . .                                                                   | 44       |
| T . . . . . GTTPGIIVTGPDGKF . . . . . LQVI . . . . .                                                                   | 42       |
| GPEG . . . . . TI . . . . . RGPSGRPIQIIP . . . . .                                                                     | 43       |
| TTT . . . . . SLRTVKGPDGEKVLLVP . . . . . NGPTPGTIVTGPGGNI . . . . . FRFI . . . . .                                    | 62       |
| PTVGP.Q . . . . . RPTIRGPDGRPLIIVP . . . . . GGPTPGAVTGPDGNI . . . . . KRII . . . . .                                  | 65       |
| GPQG . . . . . VV . . . . . KGPGGHPIQFVPGGRGSTTQAVTDDDDNL . . . . . DRII . . . . .                                     | 62       |
| APNPP.SP . . . . . LNPPGEP . . . . . IAPFGPGNSPNSP.QSPGNYPGSAFQ . . . . . F . . . . . PGFPGAPGSNGPIGYIDFSQ . . . . .   | 85       |
| PPNPP.PSGP . . . . . LNPENEP . . . . . IPPFGPGNSPNSP.QTPGNYPGSAFQ . . . . . F . . . . . PGFPDAPGSNGPIGYIDFSQ . . . . . | 85       |
| PPNPL.FSDP . . . . . LNPVGKP . . . . . AAPFGPGNIPNSP.YSPGNYPGSAFQ . . . . . F . . . . . PGFPGAPGSNGPIGYIDFSQ . . . . . | 85       |
| SPNPQ.SSGP . . . . . QYPGGGP . . . . . GGPPFGPGNSPNSPGQSPGNYPGSAFQ . . . . . F . . . . . PGFPYAPGSYGPIDYDFNH . . . . . | 87       |
| PPRG.SGAY . . . . . TLQGPSGEVIELIP . . . . .                                                                           | 46       |
| GPVG . . . . . RTAIKGGNQPILLEP . . . . . EGSTPGTIVTGPDGKI . . . . . SKIV . . . . .                                     | 62       |
| APLG . . . . . AV . . . . . KDPDSRPIEFFICGPGNRPEAVTAPDGNP . . . . . AY.I . . . . .                                     | 61       |
| PDGPG.SPEGTPGS . . . . . PNGPGRQKP . . . . . AIGG . . . . .                                                            | 49       |
| GPRG . . . . . IV . . . . . TGPKGQPIQIAPGAPGKSPKAVTSPEGNL . . . . . AKIL . . . . .                                     | 62       |
| PKRPS.YVPVITTTTVP . . . . . GAPISIVPA . . . . .                                                                        | 50       |

Ncruc 58532and13532 AgSp1 nt  
Ncruc 58532and13532 AgSp1 rr 15  
Ncruc 58532and13532 AgSp1 sy  
Ncruc 58532and13532 AgSp1 rr 14  
Ncruc 58532and13532 AgSp1 rr 7  
Ncruc 58532and13532 AgSp1 rr 12  
Ncruc 58532and13532 AgSp1 serine rm 93  
Ncruc 58532and13532 AgSp1 glycine rm 6  
Ncruc 58532and13532 AgSp1 glycine rm 1  
Ncruc 58532and13532 AgSp1 glycine rm 2  
Ncruc 58532and13532 AgSp1 glycine rm 3  
Ncruc 58532and13532 AgSp1 glycine rm 4  
Ncruc 58532and13532 AgSp1 glycine rm 5  
Ncruc 58532and13532 AgSp1 tr 1  
Ncruc 58532and13532 AgSp1 tr 3  
Ncruc 58532and13532 AgSp1 serine rm 92  
Ncruc 58532and13532 AgSp1 serine rm 91  
Ncruc 58532and13532 AgSp1 serine rm 88  
Ncruc 58532and13532 AgSp1 serine rm 89  
Ncruc 58532and13532 AgSp1 serine rm 90  
Ncruc 58532and13532 AgSp1 serine rm 87  
Ncruc 58532and13532 AgSp1 rr 1  
Ncruc 58532and13532 AgSp1 rr 11  
Ncruc 58532and13532 AgSp1 rr 2  
Ncruc 58532and13532 AgSp1 rr 6  
Ncruc 58532and13532 AgSp1 rr 9  
Ncruc 58532and13532 AgSp1 rr 3  
Ncruc 58532and13532 AgSp1 rr 13  
Mlaby 7048 AgSp1 cr  
Aarg AgSp1 FC511 cr  
Mhut 1271 AgSp1 cr  
Ncruc 58532and13532 AgSp1 ct  
Ncruc 58532and13532 AgSp1 rr 8  
Ncruc 58532and13532 AgSp1 rr 5  
Ncruc 58532and13532 AgSp1 rr 10  
Ncruc 58532and13532 AgSp1 rr 16  
Ncruc 58532and13532 AgSp1 rr 4  
Ncruc 58532and13532 AgSp1 tr 2  
consensus

[illegible]

logo

|           |               |       |                                                                                      |     |
|-----------|---------------|-------|--------------------------------------------------------------------------------------|-----|
|           |               |       | EVYPGYEGSPGSGGGQEPGGFLSFQLPKDYTDKLSSTFSFQQQLNQLPLFPQQNINLGAIPKQQLQGLPGFEGTYDNLRLANFG |     |
| Ncruc     | 58532and13532 | AgSp1 | nt                                                                                   | 7   |
| Ncruc     | 58532and13532 | AgSp1 | rr 15                                                                                | 45  |
| Ncruc     | 58532and13532 | AgSp1 | sy                                                                                   | 617 |
| Ncruc     | 58532and13532 | AgSp1 | rr 14                                                                                | 61  |
| Ncruc     | 58532and13532 | AgSp1 | rr 7                                                                                 | 45  |
| Ncruc     | 58532and13532 | AgSp1 | rr 12                                                                                | 45  |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 93                                                                         | 99  |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 6                                                                         | 59  |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 1                                                                         | 103 |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 2                                                                         | 103 |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 3                                                                         | 103 |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 4                                                                         | 103 |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 5                                                                         | 103 |
| Ncruc     | 58532and13532 | AgSp1 | tr 1                                                                                 | 45  |
| Ncruc     | 58532and13532 | AgSp1 | tr 3                                                                                 | 45  |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 92                                                                         | 96  |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 91                                                                         | 97  |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 88                                                                         | 98  |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 89                                                                         | 98  |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 90                                                                         | 109 |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 87                                                                         | 81  |
| Ncruc     | 58532and13532 | AgSp1 | rr 1                                                                                 | 48  |
| Ncruc     | 58532and13532 | AgSp1 | rr 11                                                                                | 44  |
| Ncruc     | 58532and13532 | AgSp1 | rr 2                                                                                 | 68  |
| Ncruc     | 58532and13532 | AgSp1 | rr 6                                                                                 | 43  |
| Ncruc     | 58532and13532 | AgSp1 | rr 9                                                                                 | 88  |
| Ncruc     | 58532and13532 | AgSp1 | rr 3                                                                                 | 91  |
| Ncruc     | 58532and13532 | AgSp1 | rr 13                                                                                | 89  |
| Mlaby     | 7048          | AgSp1 | cr                                                                                   | 254 |
| Aarg      | AgSp1         | FC511 | cr                                                                                   | 255 |
| Mhut      | 1271          | AgSp1 | cr                                                                                   | 255 |
| Ncruc     | 58532and13532 | AgSp1 | ct                                                                                   | 257 |
| Ncruc     | 58532and13532 | AgSp1 | rr 8                                                                                 | 68  |
| Ncruc     | 58532and13532 | AgSp1 | rr 5                                                                                 | 88  |
| Ncruc     | 58532and13532 | AgSp1 | rr 10                                                                                | 87  |
| Ncruc     | 58532and13532 | AgSp1 | rr 16                                                                                | 76  |
| Ncruc     | 58532and13532 | AgSp1 | rr 4                                                                                 | 88  |
| Ncruc     | 58532and13532 | AgSp1 | tr 2                                                                                 | 51  |
| consensus |               |       |                                                                                      |     |

logo

|           |               |                    |                                                                                          |     |
|-----------|---------------|--------------------|------------------------------------------------------------------------------------------|-----|
|           |               |                    | DNAQPIGGVFYLPPELVRLASYLPVGSFGNGPGTINQDGGFGDPFNFPGLNAGPGYICDYPDNNGDAATGRSQNLGGEIKGNHGPS   |     |
| Ncruc     | 58532and13532 | AgSp1 nt           | .....IV.....                                                                             | 9   |
| Ncruc     | 58532and13532 | AgSp1 rr 15        | .....                                                                                    | 45  |
| Ncruc     | 58532and13532 | AgSp1 sy           | .....                                                                                    | 617 |
| Ncruc     | 58532and13532 | AgSp1 rr 14        | .....                                                                                    | 61  |
| Ncruc     | 58532and13532 | AgSp1 rr 7         | .....                                                                                    | 45  |
| Ncruc     | 58532and13532 | AgSp1 rr 12        | .....                                                                                    | 45  |
| Ncruc     | 58532and13532 | AgSp1 serine rm 93 | .....                                                                                    | 99  |
| Ncruc     | 58532and13532 | AgSp1 glycine rm 6 | .....                                                                                    | 59  |
| Ncruc     | 58532and13532 | AgSp1 glycine rm 1 | .....                                                                                    | 103 |
| Ncruc     | 58532and13532 | AgSp1 glycine rm 2 | .....                                                                                    | 103 |
| Ncruc     | 58532and13532 | AgSp1 glycine rm 3 | .....                                                                                    | 103 |
| Ncruc     | 58532and13532 | AgSp1 glycine rm 4 | .....                                                                                    | 103 |
| Ncruc     | 58532and13532 | AgSp1 glycine rm 5 | .....                                                                                    | 103 |
| Ncruc     | 58532and13532 | AgSp1 tr 1         | .....                                                                                    | 45  |
| Ncruc     | 58532and13532 | AgSp1 tr 3         | .....                                                                                    | 45  |
| Ncruc     | 58532and13532 | AgSp1 serine rm 92 | .....                                                                                    | 96  |
| Ncruc     | 58532and13532 | AgSp1 serine rm 91 | .....                                                                                    | 97  |
| Ncruc     | 58532and13532 | AgSp1 serine rm 88 | .....                                                                                    | 98  |
| Ncruc     | 58532and13532 | AgSp1 serine rm 89 | .....                                                                                    | 98  |
| Ncruc     | 58532and13532 | AgSp1 serine rm 90 | .....                                                                                    | 109 |
| Ncruc     | 58532and13532 | AgSp1 serine rm 87 | .....                                                                                    | 81  |
| Ncruc     | 58532and13532 | AgSp1 rr 1         | .....                                                                                    | 48  |
| Ncruc     | 58532and13532 | AgSp1 rr 11        | .....                                                                                    | 44  |
| Ncruc     | 58532and13532 | AgSp1 rr 2         | .....                                                                                    | 68  |
| Ncruc     | 58532and13532 | AgSp1 rr 6         | .....                                                                                    | 43  |
| Ncruc     | 58532and13532 | AgSp1 rr 9         | .....                                                                                    | 88  |
| Ncruc     | 58532and13532 | AgSp1 rr 3         | .....                                                                                    | 91  |
| Ncruc     | 58532and13532 | AgSp1 rr 13        | .....                                                                                    | 89  |
| Mlaby     | 7048          | AgSp1 cr           | DSSQPTGGVFYLPPELVRLASYLPVGSFGNGPGTINQDGGFGDPFNFPGLNAGPGYICDYPDNDDTTAGGSQDLGGEIKGNHGPS    | 332 |
| Aarg      | AgSp1 FC511   | cr                 | NNNQPVGGVFYLPPELVRLISYLPVGSFGNGPGTINQDGGFGDPFNFPGLNAGPGYICDYPDNNGDAATGRSQNLGGEIKGNHGPS   | 340 |
| Mhut      | 1271          | AgSp1 cr           | DNSQPTGGVFYLPPELVRLIGYLPVVSSSKDSEIINKNGGFSNPFNFPGLNAGPGYICDYPDNNGDATTEGSQDLGGEIEGYGNNGPF | 340 |
| Ncruc     | 58532and13532 | AgSp1 ct           | DNGQPTGGVFYLPPELVRLMSYLPVGSFGKGPGTINHGGFDDPFNFPGLNAGPGSICDYPDNNGDATDGGSQDLGGEVKGDNNGPA   | 342 |
| Ncruc     | 58532and13532 | AgSp1 rr 8         | .....                                                                                    | 68  |
| Ncruc     | 58532and13532 | AgSp1 rr 5         | .....                                                                                    | 88  |
| Ncruc     | 58532and13532 | AgSp1 rr 10        | .....                                                                                    | 87  |
| Ncruc     | 58532and13532 | AgSp1 rr 16        | .....                                                                                    | 76  |
| Ncruc     | 58532and13532 | AgSp1 rr 4         | .....                                                                                    | 88  |
| Ncruc     | 58532and13532 | AgSp1 tr 2         | .....                                                                                    | 51  |
| consensus |               |                    |                                                                                          |     |

| Accession           | Protein Name       | Length                                                         | Score |
|---------------------|--------------------|----------------------------------------------------------------|-------|
| Ncruc 58532and13532 | AgSp1 nt           | .....                                                          | 9     |
| Ncruc 58532and13532 | AgSp1 rr 15        | .....                                                          | 45    |
| Ncruc 58532and13532 | AgSp1 sy           | .....                                                          | 617   |
| Ncruc 58532and13532 | AgSp1 rr 14        | .....                                                          | 61    |
| Ncruc 58532and13532 | AgSp1 rr 7         | .....                                                          | 45    |
| Ncruc 58532and13532 | AgSp1 rr 12        | .....                                                          | 45    |
| Ncruc 58532and13532 | AgSp1 serine rm 93 | .....                                                          | 99    |
| Ncruc 58532and13532 | AgSp1 glycine rm 6 | .....                                                          | 59    |
| Ncruc 58532and13532 | AgSp1 glycine rm 1 | .....                                                          | 103   |
| Ncruc 58532and13532 | AgSp1 glycine rm 2 | .....                                                          | 103   |
| Ncruc 58532and13532 | AgSp1 glycine rm 3 | .....                                                          | 103   |
| Ncruc 58532and13532 | AgSp1 glycine rm 4 | .....                                                          | 103   |
| Ncruc 58532and13532 | AgSp1 glycine rm 5 | .....                                                          | 103   |
| Ncruc 58532and13532 | AgSp1 tr 1         | .....                                                          | 45    |
| Ncruc 58532and13532 | AgSp1 tr 3         | .....                                                          | 45    |
| Ncruc 58532and13532 | AgSp1 serine rm 92 | .....                                                          | 96    |
| Ncruc 58532and13532 | AgSp1 serine rm 91 | .....                                                          | 97    |
| Ncruc 58532and13532 | AgSp1 serine rm 88 | .....                                                          | 98    |
| Ncruc 58532and13532 | AgSp1 serine rm 89 | .....                                                          | 98    |
| Ncruc 58532and13532 | AgSp1 serine rm 90 | .....                                                          | 109   |
| Ncruc 58532and13532 | AgSp1 serine rm 87 | .....                                                          | 81    |
| Ncruc 58532and13532 | AgSp1 rr 1         | .....                                                          | 48    |
| Ncruc 58532and13532 | AgSp1 rr 11        | .....                                                          | 44    |
| Ncruc 58532and13532 | AgSp1 rr 2         | .....                                                          | 68    |
| Ncruc 58532and13532 | AgSp1 rr 6         | .....                                                          | 43    |
| Ncruc 58532and13532 | AgSp1 rr 9         | .....                                                          | 88    |
| Ncruc 58532and13532 | AgSp1 rr 3         | .....                                                          | 91    |
| Ncruc 58532and13532 | AgSp1 rr 13        | .....                                                          | 89    |
| Mlaby 7048          | AgSp1 cr           | GDVADAAPNLGGSAPGNDLGGAAPGNTLGG                                 | 371   |
| Aarg                | AgSp1 FC511 cr     | GDVADAAPNLDLGGAPGNTLGG                                         | 351   |
| Mhut 1271           | AgSp1 cr           | EDVADAAPNLDLGGAPGNTLGG                                         | 351   |
| Ncruc 58532and13532 | AgSp1 ct           | GDVADAAPDLGAPAGNDLGGAPDL..GAPAGNDLGGAPDLGAPAGNDLGGAPDLGAPAGNDL | 425   |
| Ncruc 58532and13532 | AgSp1 rr 8         | .....                                                          | 68    |
| Ncruc 58532and13532 | AgSp1 rr 5         | .....                                                          | 88    |
| Ncruc 58532and13532 | AgSp1 rr 10        | .....                                                          | 87    |
| Ncruc 58532and13532 | AgSp1 rr 16        | .....                                                          | 76    |
| Ncruc 58532and13532 | AgSp1 rr 4         | .....                                                          | 88    |
| Ncruc 58532and13532 | AgSp1 tr 2         | .....                                                          | 51    |
| consensus           |                    |                                                                |       |

logo

|                                        |                                                                                       |     |
|----------------------------------------|---------------------------------------------------------------------------------------|-----|
| Ncruc 58532and13532 AgSp1 nt           | .....LLCLVGTQLTKAYPQQNYQPGIGIAQDDPLLEDGTGENFASAFQSLNCGAFTH EE.AQEYKEIIQVLTR.....      | 79  |
| Ncruc 58532and13532 AgSp1 rr 15        | .....                                                                                 | 45  |
| Ncruc 58532and13532 AgSp1 sy           | .....                                                                                 | 617 |
| Ncruc 58532and13532 AgSp1 rr 14        | .....                                                                                 | 61  |
| Ncruc 58532and13532 AgSp1 rr 7         | .....                                                                                 | 45  |
| Ncruc 58532and13532 AgSp1 rr 12        | .....                                                                                 | 45  |
| Ncruc 58532and13532 AgSp1 serine rm 93 | .....                                                                                 | 99  |
| Ncruc 58532and13532 AgSp1 glycine rm 6 | .....                                                                                 | 59  |
| Ncruc 58532and13532 AgSp1 glycine rm 1 | .....                                                                                 | 103 |
| Ncruc 58532and13532 AgSp1 glycine rm 2 | .....                                                                                 | 103 |
| Ncruc 58532and13532 AgSp1 glycine rm 3 | .....                                                                                 | 103 |
| Ncruc 58532and13532 AgSp1 glycine rm 4 | .....                                                                                 | 103 |
| Ncruc 58532and13532 AgSp1 glycine rm 5 | .....                                                                                 | 103 |
| Ncruc 58532and13532 AgSp1 tr 1         | .....                                                                                 | 45  |
| Ncruc 58532and13532 AgSp1 tr 3         | .....                                                                                 | 45  |
| Ncruc 58532and13532 AgSp1 serine rm 92 | .....                                                                                 | 96  |
| Ncruc 58532and13532 AgSp1 serine rm 91 | .....                                                                                 | 97  |
| Ncruc 58532and13532 AgSp1 serine rm 88 | .....                                                                                 | 98  |
| Ncruc 58532and13532 AgSp1 serine rm 89 | .....                                                                                 | 98  |
| Ncruc 58532and13532 AgSp1 serine rm 90 | .....                                                                                 | 109 |
| Ncruc 58532and13532 AgSp1 serine rm 87 | .....                                                                                 | 81  |
| Ncruc 58532and13532 AgSp1 rr 1         | .....                                                                                 | 48  |
| Ncruc 58532and13532 AgSp1 rr 11        | .....                                                                                 | 44  |
| Ncruc 58532and13532 AgSp1 rr 2         | .....                                                                                 | 68  |
| Ncruc 58532and13532 AgSp1 rr 6         | .....                                                                                 | 43  |
| Ncruc 58532and13532 AgSp1 rr 9         | .....                                                                                 | 88  |
| Ncruc 58532and13532 AgSp1 rr 3         | .....                                                                                 | 91  |
| Ncruc 58532and13532 AgSp1 rr 13        | .....                                                                                 | 89  |
| Mlaby 7048 AgSp1 cr                    | .....SAPGNAPGGSAPGNNMASNTLVGPED....EGCQDNVPEAFSKARQALA.....                           | 416 |
| Aarg AgSp1 FC511 cr                    | .....LGGAAPQSDLASPTPVAPTD....QDCEEDVHGAFSKARQALM.....                                 | 390 |
| Mhut 1271 AgSp1 cr                     | .....LEGVASQSNLGPNTLVAPEV....KDCQDSVPEAFRKARHALT.....                                 | 390 |
| Ncruc 58532and13532 AgSp1 ct           | GGAAADLGAPAPAPGNDLGATDPPSDQGSNTLVAPDD....EDCQDNVPEAFHKARQALANVESSVGQQTVSQLMQAIISGINIP | 506 |
| Ncruc 58532and13532 AgSp1 rr 8         | .....                                                                                 | 68  |
| Ncruc 58532and13532 AgSp1 rr 5         | .....                                                                                 | 88  |
| Ncruc 58532and13532 AgSp1 rr 10        | .....                                                                                 | 87  |
| Ncruc 58532and13532 AgSp1 rr 16        | .....                                                                                 | 76  |
| Ncruc 58532and13532 AgSp1 rr 4         | .....                                                                                 | 88  |
| Ncruc 58532and13532 AgSp1 tr 2         | .....                                                                                 | 51  |
| consensus                              |                                                                                       |     |

logo

|           |               |       |              |                                                                                      |     |
|-----------|---------------|-------|--------------|--------------------------------------------------------------------------------------|-----|
| Ncruc     | 58532and13532 | AgSp1 | nt           | .AFGSFPDLQKAPPGVLKATAAGFASAVAERTASDIEKGELFRKTRCVIQALREA...FIVTTGV..TNPYFISEVERLIEVFY | 157 |
| Ncruc     | 58532and13532 | AgSp1 | rr 15        | .....                                                                                | 45  |
| Ncruc     | 58532and13532 | AgSp1 | sy           | .....                                                                                | 617 |
| Ncruc     | 58532and13532 | AgSp1 | rr 14        | .....                                                                                | 61  |
| Ncruc     | 58532and13532 | AgSp1 | rr 7         | .....                                                                                | 45  |
| Ncruc     | 58532and13532 | AgSp1 | rr 12        | .....                                                                                | 45  |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 93 | .....                                                                                | 99  |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 6 | .....                                                                                | 59  |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 1 | .....                                                                                | 103 |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 2 | .....                                                                                | 103 |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 3 | .....                                                                                | 103 |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 4 | .....                                                                                | 103 |
| Ncruc     | 58532and13532 | AgSp1 | glycine rm 5 | .....                                                                                | 103 |
| Ncruc     | 58532and13532 | AgSp1 | tr 1         | .....                                                                                | 45  |
| Ncruc     | 58532and13532 | AgSp1 | tr 3         | .....                                                                                | 45  |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 92 | .....                                                                                | 96  |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 91 | .....                                                                                | 97  |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 88 | .....                                                                                | 98  |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 89 | .....                                                                                | 98  |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 90 | .....                                                                                | 109 |
| Ncruc     | 58532and13532 | AgSp1 | serine rm 87 | .....                                                                                | 81  |
| Ncruc     | 58532and13532 | AgSp1 | rr 1         | .....                                                                                | 48  |
| Ncruc     | 58532and13532 | AgSp1 | rr 11        | .....                                                                                | 44  |
| Ncruc     | 58532and13532 | AgSp1 | rr 2         | .....                                                                                | 68  |
| Ncruc     | 58532and13532 | AgSp1 | rr 6         | .....                                                                                | 43  |
| Ncruc     | 58532and13532 | AgSp1 | rr 9         | .....                                                                                | 88  |
| Ncruc     | 58532and13532 | AgSp1 | rr 3         | .....                                                                                | 91  |
| Ncruc     | 58532and13532 | AgSp1 | rr 13        | .....                                                                                | 89  |
| Mlaby     | 7048          | AgSp1 | cr           | .....                                                                                | 416 |
| Aarg      |               | AgSp1 | FC511 cr     | .....                                                                                | 390 |
| Mhut      | 1271          | AgSp1 | cr           | .....                                                                                | 390 |
| Ncruc     | 58532and13532 | AgSp1 | ct           | GNVVDYNDFFNQLSSLLSQVRSGTDSPTKQ....V.DSVLMEALIAALEALNSAKVTGFRNDVSVANDLPVYTSF...LNEILY | 582 |
| Ncruc     | 58532and13532 | AgSp1 | rr 8         | .....                                                                                | 68  |
| Ncruc     | 58532and13532 | AgSp1 | rr 5         | .....                                                                                | 88  |
| Ncruc     | 58532and13532 | AgSp1 | rr 10        | .....                                                                                | 87  |
| Ncruc     | 58532and13532 | AgSp1 | rr 16        | .....                                                                                | 76  |
| Ncruc     | 58532and13532 | AgSp1 | rr 4         | .....                                                                                | 88  |
| Ncruc     | 58532and13532 | AgSp1 | tr 2         | .....                                                                                | 51  |
| consensus |               |       |              |                                                                                      |     |

 non-conserved  
 similar  
 ≥ 50% conserved